

Data Structure and algorithms (LAB)

Lab Task

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Problem 1

```
#include <iostream>
```

```
using namespace std;
```

```
struct Node
```

```
{
```

```
    int data;
```

```
    Node *next;
```

```
};
```

```
Node *head = NULL;
```

```
int main()
```

```
{
```

```
    Node *currentNode = NULL;
```

```
    for (int i = 1; i <= 10; i += 2)
```

```
    {
```

```
        Node *newNode = new Node;
```

```
        newNode->data = i * 10;
```

```
        newNode->next = NULL;
```

```
    if (head == NULL)
    {
        head = newNode;
        currentNode = newNode;
    }
    else
    {
        currentNode->next = newNode;
        currentNode = newNode;
    }
}

int i = 1;

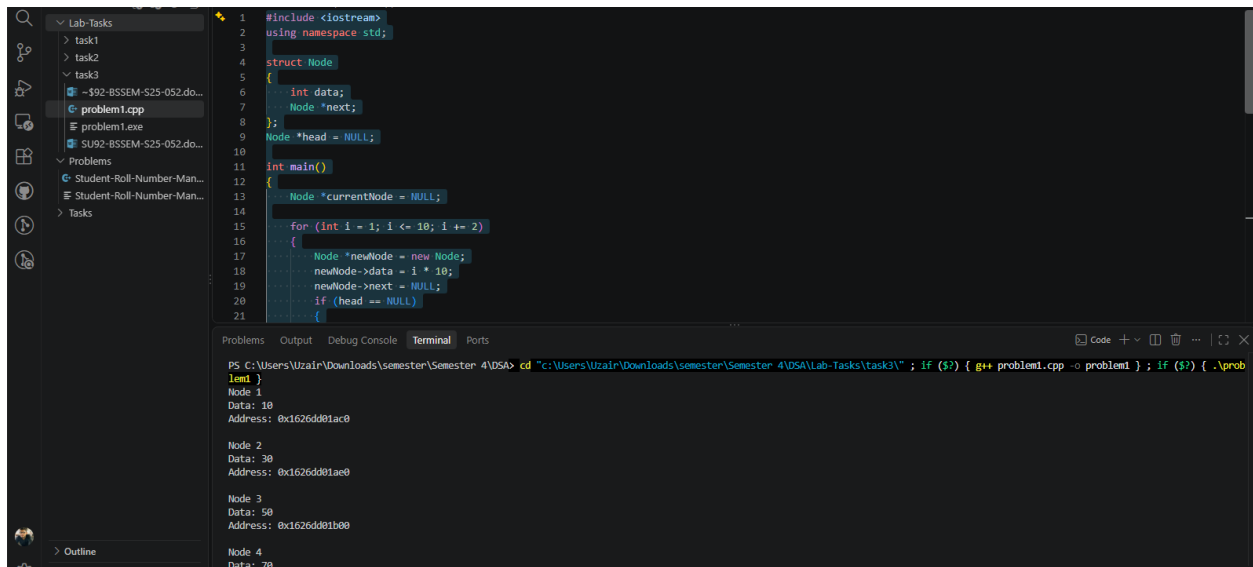
currentNode = head;

while (currentNode != NULL)
{
    cout << "Node " << i << "\nData: " << currentNode->data << "\nAddress: " << currentNode->next << "\n"
        << endl;

    currentNode = currentNode->next;

    i++;
}

return 0;
}
```



Problem 2

```
#include <iostream>
```

```
using namespace std;
```

```
struct Node
```

```
{
```

```
    int data;
```

```
    Node *next;
```

```
};
```

```
Node *head = NULL;
```

```
int main()
```

```
{
```

```
    Node *currentNode = NULL;
```

```
for (int i = 2; i <= 10; i += 2)
{
    Node *newNode = new Node;

    newNode->data = i * 10;

    newNode->next = NULL;

    if (head == NULL)
    {
        head = newNode;

        currentNode = newNode;
    }
    else
    {
        currentNode->next = newNode;

        currentNode = newNode;
    }
}

currentNode = head;

while (currentNode != NULL)
{
    cout << currentNode->data << " --> ";

    currentNode = currentNode->next;
}

cout << "NULL" << endl;
```

```

Node *newNode = new Node;

newNode->data = 10;

newNode->next = head;

head = newNode;

currentNode = head;

while (currentNode != NULL)
{
    cout << currentNode->data << " --> ";

    currentNode = currentNode->next;
}

cout << "NULL" << endl;

return 0;
}

```

```

24     }
25     }
26     }
27     {
28         currentNode->next = newNode;
29         currentNode = newNode;
30     }
31 }
32
33 currentNode = head;
34
35 while (currentNode != NULL)
36 {
37     cout << currentNode->data << " --> ";
38     currentNode = currentNode->next;
39 }
40
41 cout << "NULL" << endl;
42
43 Node *newNode = new Node;
44 newNode->data = 10;
45 newNode->next = head;

```

Problems Output Debug Console **Terminal** Ports

```

PS C:\Users\Uzair\Downloads\semester\Semester 4\DSA> cd "C:\Users\Uzair\Downloads\semester\Semester 4\DSA\Lab-Tasks\task3" ; if ($?) { g++ problem2.cpp -o problem2 } ; if ($?) { .\problem2 }
20 --> 40 --> 60 --> 80 --> 100 --> NULL
10 --> 20 --> 40 --> 60 --> 80 --> 100 --> NULL
PS C:\Users\Uzair\Downloads\semester\Semester 4\DSA\Lab-Tasks\task3>

```